Math 2C03 - Assignment 2

Fall 2026

This is one generated instance of the assignment. Your Childsmath questions may have different numerical values or choice orders. Submit your answers in Childsmath.

Question 1

Consider the following autonomous first order differential equation.

$$\frac{dy}{dx} = (y-2)(y+7)^2.$$

The critical points are $y = 2$ and $y = -7$. Classify these critical points (in the given order) as asymptotically stable, unstable, or semi-stable.

(A) asymptotically stable and asymptotically stable	(B) asymptotically stable and semi-stable
(C) semi-stable and unstable	(D) unstable and asymptotically stable
(E) semi-stable and semi-stable	(F) asymptotically stable and unstable
(G) semi-stable and asymptotically stable	(H) unstable and unstable
(I) unstable and semi-stable	

Question 2

Solve the following initial value problem.

$$\cos^2 x \sin x \frac{dy}{dx} + (\cos^3 x)y = 4, \qquad y(\pi/3) = 3.$$

Question 3

Consider the differential equation

$$(4xy^2 + 3y + Ax^2y) + (4x^2y + Bx + 2x^3)\frac{dy}{dx} = 0.$$

(a) Find the values of A and B which make this differential equation exact. Enter the values of A and B (in that order), separated with a comma.

(b) Consider the differential equation above with A and B replaced by the values found in part (a). If the solution of the differential equation above satisfying the initial condition $y(1) = 1$ is written in implicit form as the equation $f(x, y) = 7$, compute the function $f(x, y)$.

Question 4

If the solution of the initial value problem

$$(3x^2 + y)\,dx + (x^2y - x)\,dy = 0, \qquad y(2) = 12$$

is written in implicit form as the equation $F(x, y) = 72$, compute the function $F(x, y)$.
Hint: Find an integrating factor.

Question 5

Consider the differential equation

$$(6x^n + 2xy)y' = 3x^n + 6xy + 2y^n.$$

(a) For which value of n is the first-order (nonlinear) differential equation above homogeneous?

(b) For the value of n found in part (a), use the method of homogeneous equations to find in explicit form the unique solution of the differential equation above that satisfies the initial condition $y(1) = -6$.

Question 6

Solve the initial-value problem

$$8y' + (1 + 3x^2)y = \frac{e^{-x^3}}{y^7}, \qquad y(0) = 1.$$

Note that this is a Bernoulli differential equation.

Question 7

Solve the following differential equation,

$$(2tx + 3t^2)x' = x^2,$$

with initial conditions

(a) $x(1) = 5$;

(b) $x(1) = -3$.

Question 8

Solve the following initial value problem.

$$tx' - (8 + t)x = t^{-7}x^2, \qquad x(1) = \frac{1}{5}.$$